

Student ID: 202618015

Course: DS602

1. Project Overview

This project involves the execution of an end-to-end data science workflow, transitioning from raw tabular data to an interactive, user-centric web application. Utilizing the **Medical Insurance Costs** dataset (1,338 observations), the project applies exploratory data analysis (EDA), rigorous hypothesis testing, and multiple linear regression to uncover the primary drivers of healthcare charges. The final deliverable is a live Streamlit dashboard deployed on the cloud, enabling users to interact with statistical models in real time.

2. Purpose & Objectives

The primary purpose of this project is to bridge the gap between formal mathematical modeling and real-world software delivery. Specific objectives include:

- Translating static code into an interactive tool for non-technical stakeholders.
- Evaluating parametric and non-parametric statistical hypothesis tests based on data distribution assumptions.
- Fitting an Ordinary Least Squares (OLS) regression model to predict insurance costs based on demographic and lifestyle factors.
- Validating the reliability of the model through Gauss-Markov diagnostic checks.

3. Key Definitions & Terminology

- Null Hypothesis (H_0): The baseline assumption that there is no significant difference between groups or no relationship between variables. We test against a significance level of $\alpha = 0.05$.
- p-value: The probability of obtaining the observed results assuming the null hypothesis is true. A p-value < 0.05 indicates statistical significance.
- Ordinary Least Squares (OLS): A method for estimating the unknown parameters in a linear regression model by minimizing the sum of the squared differences (residuals) between the observed and predicted values.
- Homoscedasticity: The assumption that the variance of the residuals (errors) is constant across all levels of the independent variables.
- Multicollinearity: A phenomenon where independent variables in a regression model are highly correlated, calculated using the Variance Inflation Factor (VIF). Values > 5 indicate problematic collinearity.

4. Mathematical Formulas

1. Multiple Linear Regression Model The core predictive model used in this dashboard is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

(Where Y is the predicted medical charge, β_0 is the intercept, β_k are the coefficients for features like age or BMI, and ε represents the error term).

2. Residuals Diagnostic checks rely on analyzing the residuals (e_i), calculated as the difference between the actual observed value (Y_i) and the predicted value ($\hat{Y}_i$):

$$e_i = Y_i - \hat{Y}_i$$

5. Methodology & Implementation

1. Exploratory Data Analysis (EDA):

- Calculated measures of central tendency (mean, median) and dispersion (standard deviation, IQR, skewness, kurtosis).
- Visualized data using histograms and scatter plots with Ordinary Least Squares trendlines to identify preliminary non-linear relationships (e.g., the interaction between BMI and smoking).

2. Hypothesis Testing:

- **Assumption Checks:** Used the Shapiro-Wilk test for normality and Levene's test for equal variance.
- **Two-Sample Tests:** Deployed the Independent t-test for normally distributed data, falling back to the non-parametric Mann-Whitney U test when the normality assumption was violated (e.g., comparing charges between smokers and non-smokers).
- **Multi-Group Tests:** Utilized One-Way ANOVA to test for variance in numerical metrics across the four geographic regions.

3. Statistical Modeling & Diagnostics:

- Categorical variables (sex, smoker, region) were preprocessed using dummy encoding.
- Fitted an OLS regression model via the statsmodels library.
- Performed Gauss-Markov diagnostic checks: Plotted Residuals vs. Fitted values to check for linearity/homoscedasticity, and generated a Q-Q plot to verify the normality of residuals.

4. Software Engineering & Deployment:

- Architected a 3-tab graphical user interface using Streamlit (app.py).
- Implemented session state and caching (@st.cache_data) for performance optimization.
- Deployed the final environment via Streamlit Community Cloud for public access.

6. Synthesis of Findings (What Was Understood)

Through the development of this dashboard and the execution of the statistical tests, several key insights were validated:

- **Smoking is the Dominant Cost Driver:** Hypothesis testing definitively rejected the null hypothesis regarding smoking status. The model coefficients confirm that being a smoker exponentially increases medical charges compared to any other single feature.
- **Age and BMI Have Positive Correlations:** As age and BMI increase, medical charges predictably rise. The interactive dashboard clearly visualizes how adjusting the BMI slider for a smoker dramatically alters the 95% prediction interval.
- **Model Validity:** The diagnostic plots (Tab 3) demonstrate that the OLS model captures the general trend effectively. Multicollinearity was successfully managed (VIF scores remained within acceptable bounds).
- **The Value of Interactivity:** By transitioning from a static Jupyter Notebook to a Streamlit app, we learned that allowing users to input their own theoretical data (e.g., changing a patient's age and instantly seeing the updated predicted cost) makes statistical modeling vastly more accessible and actionable for decision-makers.